

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2016–17

Paper: BCS-301 — Numerical Computing

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Use of calculator is allowed. All undefined symbols have their usual meanings. Each question carries **14 marks**.*

Question 1. (Compulsory — attempt all parts)

[2 × 7 = 14]

- Explain different types of errors with an example.
- If the first significant figure of a number is k and the number is correct to n significant figures, prove that the relative error is less than $\frac{1}{2k \times 10^{n-1}}$.
- Prove that the Newton–Raphson process has second-order (quadratic) convergence.
- Given that the equation $x^3 - 7 = 69$ has a root between 5 and 8, apply the method of Regula-Falsi (False Position) to determine it.
- Define Finite Differences. Explain Forward, Backward, and Central differences with examples.
- Evaluate $\Delta \tan^{-1} x$.
- State the conditions under which the Gauss–Seidel iteration method converges.

Question 2.

[6 + 8 = 14]

- Find a real root of the equation $x^3 - 2x - 5 = 0$ using the Bisection Method, correct to three decimal places.
- From the following table, estimate the number of students who obtained marks between 40 and 45 using Newton’s Forward Interpolation Formula:

Marks	30–40	40–50	50–60	60–70	70–80
No. of Students	31	42	51	35	31

Question 3.

[6 + 8 = 14]

- Apply the Gauss Elimination method to solve the following system of equations, correct to three decimal places:

$$x + y + z = 9, \quad 2x - 3y + 4z = 13, \quad 3x + 4y + 5z = 40$$

- Apply the Gauss–Jordan method to solve the following system, correct to three decimal places:

$$x + 4y - z = -5, \quad x + y - 6z = -12, \quad 3x - y - z = 4$$

Question 4. The table gives the distance y (in nautical miles) of the visible horizon for given heights x (in feet) above the earth's surface: [6 + 8 = 14]

x (ft)	100	150	200	250	300	350	400
y (mi.)	10.63	13.03	15.04	16.81	18.42	19.90	21.27

Find the values of y when:

- (a) $x = 218$ ft
- (b) $x = 410$ ft

Question 5. Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using the following rules and compare with the actual value: [3 + 4 + 4 + 3 = 14]

- (a) Trapezoidal Rule
- (b) Simpson's $\frac{1}{3}$ Rule
- (c) Simpson's $\frac{3}{8}$ Rule
- (d) Weddle's Rule

Question 6. Using the Runge-Kutta method of fourth order, solve $\frac{dy}{dx} = x + y$ with $y(0) = 1$ at $x = 0.2$ and $x = 0.4$. [14]

Question 7. [6 + 8 = 14]

- (a) Apply Milne's method to find a solution of the differential equation $y' = x - y^2$ in the range $0 \leq x \leq 1$ for the boundary condition $y = 0$ at $x = 0$.
- (b) Given $\frac{dx}{dy} = x^2(1 + y)$ and $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$, evaluate $y(1.4)$ by the Adams-Bashforth method.